

Lecture 6: Theories of unconstrained optimization (part 1/2)

MA3236 Non-Linear Programming

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Quick review for Topic 1

- ① **Existence of global:** continuity + compactness / continuity + coercivity.
- ② **Local versus global:** a local optimizer only wins nearby.
 $\Rightarrow$ Define and justify what we are solving for abstractly.

Quick review for Topic 1

- 1 **Existence of global:** continuity + compactness / continuity + coercivity.
- 2 **Local versus global:** a local optimizer only wins nearby.
- 3 **Convex optimization as an outlier:**
 - Every local is global.
 - Necessary and sufficient optimality condition.

$$\mathbf{x}^* \in \arg \min_{\mathbf{x} \in C} f(\mathbf{x}) \quad \Leftrightarrow \quad \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \forall \mathbf{x} \in C.$$

If C is open (e.g., $C = \mathbf{R}^n$), $\nabla f(\mathbf{x}^*) = 0$

$\Rightarrow$ Characterize what we are solving for algebraically **in special cases**.

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If C is open (e.g., $C = \mathbf{R}^n$), $\nabla f(\mathbf{x}^*) = 0$

- ④ **Convexity tests for functions:** definition $(\mathbf{x}, \mathbf{y}, \lambda)$, first-order $(\mathbf{x}, \mathbf{y})$, second-order $(\mathbf{x})$.

$\Rightarrow$ Provide building blocks for algorithm design (to be shown later).

What survives when convexity is removed?

Consider

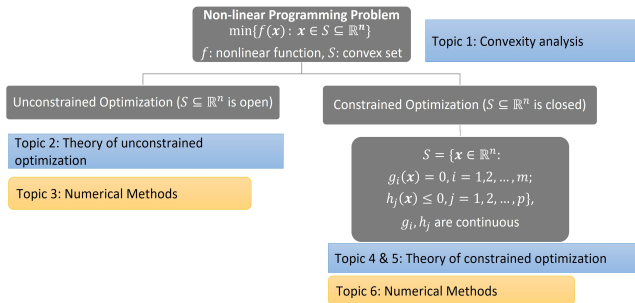
$$\min_{\mathbf{x} \in S} f(\mathbf{x}), \quad S \subseteq \mathbf{R}^n \text{ open},$$

where f is smooth but possibly nonconvex, S is possibly nonconvex.

The question for these two lectures

Is there a derivative condition that is necessary and/or sufficient for a local or global optimizer?

Topics



In these two lectures, we will study theories of **unconstrained optimization problem**.

- Necessary condition for any local minimizer to satisfy.
- Sufficient condition to tell strict local minimizers/maximizers.

First-order necessary condition: one-dimensional cases

Theorem 6.1 (Fermat's necessary condition in one dimension)

Let $\mathcal{D} \subseteq \mathbf{R}$ be open and let $\phi : \mathcal{D} \rightarrow \mathbf{R}$ be differentiable at $x^* \in \mathcal{D}$.

- If x^* is a local minimizer, then $\phi'(x^*) = 0$.
- If x^* is a local maximizer, then $\phi'(x^*) = 0$.

Proof. Suppose that x^* is a local minimizer. For all sufficiently small h ,

$$\phi(x^* + h) - \phi(x^*) \geq 0,$$

so

$$\frac{\phi(x^* + h) - \phi(x^*)}{h} \begin{cases} \geq 0, & h > 0, \\ \leq 0, & h < 0. \end{cases}$$

First-order necessary condition: one-dimensional cases

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Apply the same argument to $-\phi$ for a local maximum. □

Extension to multi-dimensional cases

Fix a direction $\mathbf{d} \in \mathbf{R}^n$ and restrict f to the line through $\mathbf{x}^*$:

$$\phi_{\mathbf{d}}(t) := f(\mathbf{x}^* + t\mathbf{d}).$$

If $\mathbf{x}^*$ is an interior local minimizer of f , then $t = 0$ is a local minimizer of every $\phi_{\mathbf{d}}$. Therefore

$$0 = \phi'_{\mathbf{d}}(0) = \nabla f(\mathbf{x}^*)^T \mathbf{d} \quad \forall \mathbf{d} \in \mathbf{R}^n.$$

The only vector orthogonal to every direction is the zero vector.

First-order necessary condition: multi-dimensional cases

Theorem 6.2 (First-order necessary condition)

Let $\mathcal{D} \subseteq \mathbf{R}^n$ be open and let $f : \mathcal{D} \rightarrow \mathbf{R}$ be differentiable at $\mathbf{x}^* \in \mathcal{D}$. If $\mathbf{x}^*$ is a local minimizer or local maximizer, then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Definition 6.3 (Stationary point)

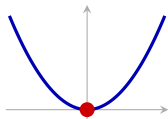
Let $\mathcal{D} \subseteq \mathbf{R}^n$ be open. A point $\mathbf{x}^* \in \mathcal{D}$ is a **stationary point** of f if

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Stationarity means zero first-order change in every direction. It does not yet say whether nearby values are larger, smaller, or mixed.

Stationarity alone allows opposite roles

At $x^* = 0$, all three functions satisfy $\phi'(0) = 0$.



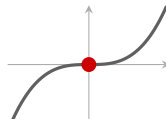
$$\phi(x) = x^2$$

Role of 0: global minimum



$$\phi(x) = -x^2$$

Role of 0: global maximum



$$\phi(x) = x^3$$

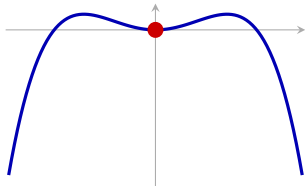
Role of 0: saddle point

Definition 6.4 (Saddle point)

A stationary point that is neither a local minimizer nor a local maximizer is a **saddle point**.

Local behavior need not determine the global role

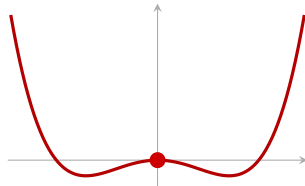
Both functions satisfy $\phi'(0) = 0$.



$$\phi(x) = x^2 - x^4$$

Near 0: $\phi(x) \geq 0$; for $|x| > 1$: $\phi(x) < 0$.

Role of 0: strict local minimum, not global



$$\phi(x) = -x^2 + x^4$$

Near 0: $\phi(x) \leq 0$; for $|x| > 1$: $\phi(x) > 0$.

Role of 0: strict local maximum, not global

The stationary set contains several possible roles

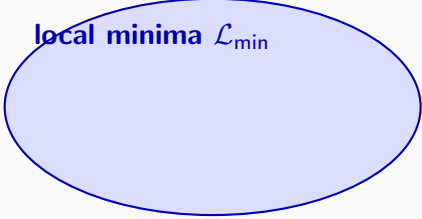
$$\mathcal{S} = \{\mathbf{x} : \nabla f(\mathbf{x}) = \mathbf{0}\}$$

For differentiable f on an open domain: $\mathcal{G}_{\min} \subseteq \mathcal{L}_{\min} \subseteq \mathcal{S}$ and $\mathcal{G}_{\max} \subseteq \mathcal{L}_{\max} \subseteq \mathcal{S}$. The diagram shows the nonconstant case, so $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} = \emptyset$. If they intersect, f is constant and all five sets equal $\mathcal{D}$.

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local minima $\mathcal{L}_{\min}$



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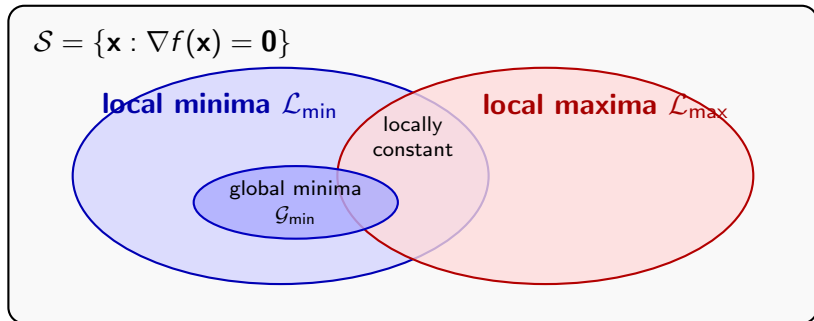
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local minima $\mathcal{L}_{\min}$

global minima
 $\mathcal{G}_{\min}$

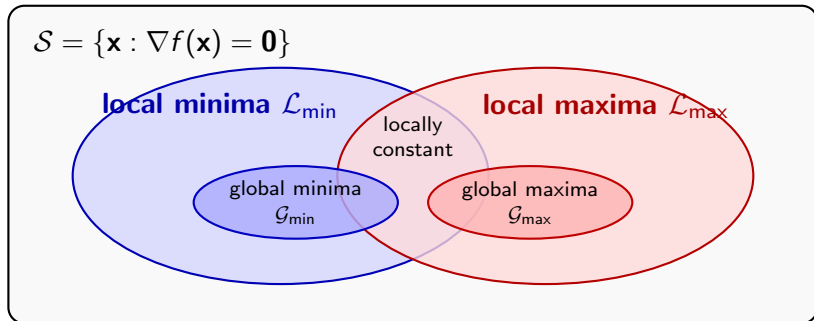
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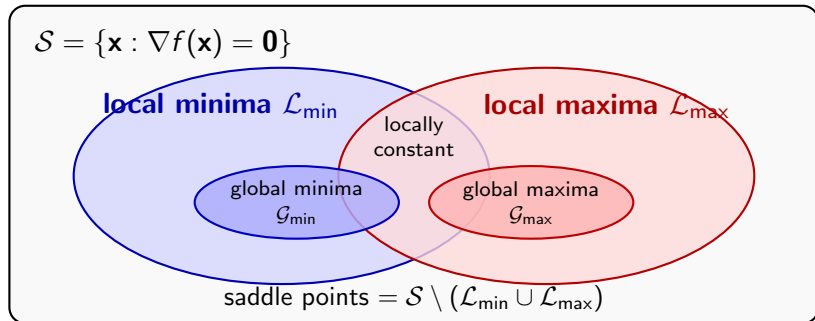
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Under convexity, stationary points are global minima

Assume that $f : \mathcal{D} \rightarrow \mathbb{R}$ is differentiable and convex on an open convex domain $\mathcal{D}$.

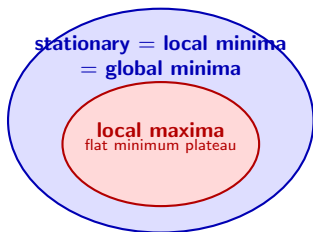


**stationary = local minima
= global minima**

saddle points = $\emptyset$; $\mathcal{L}_{\max} \subseteq \mathcal{G}_{\min}$, $\mathcal{G}_{\max} = \emptyset$.

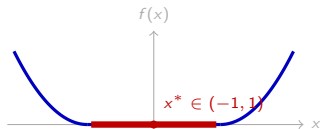
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A local maximum without global constancy

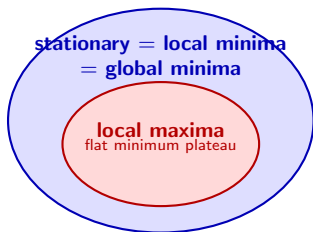


$$f(x) = (\max\{|x| - 1, 0\})^2$$

Plateau interior: global min + *non-strict* local max.

Under convexity, stationary points are global minima

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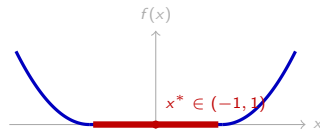


saddle points = $\emptyset$; $\mathcal{L}_{\max} \subseteq \mathcal{G}_{\min}$, $\mathcal{G}_{\max} = \emptyset$.



saddle points = $\emptyset$;
 $\mathcal{G}_{\max} \neq \emptyset$ only if f is constant, all sets coincide.

A local maximum without global constancy



$$f(x) = (\max\{|x| - 1, 0\})^2$$

Plateau interior: global min + *non-strict* local max.

Globally constant: $g(x) \equiv 0$



Every point is both a global minimum and a global maximum.

Second-order necessary condition: one-dimensional cases

Necessary conditions at an interior local optimizer

Let ϕ be twice differentiable near x^* . Then

$$x^* \text{ local minimum} \implies \phi'(x^*) = 0, \quad \phi''(x^*) \geq 0,$$

$$x^* \text{ local maximum} \implies \phi'(x^*) = 0, \quad \phi''(x^*) \leq 0.$$

Proof. The first-order condition gives $\phi'(x^*) = 0$. Taylor expansion yields

$$\phi(x^* + h) - \phi(x^*) = \frac{1}{2}\phi''(x^*)h^2 + o(h^2).$$

At a local minimum, the left-hand side is nonnegative for all sufficiently small h . Divide by $h^2 > 0$ and let $h \rightarrow 0$ to obtain $\phi''(x^*) \geq 0$. Apply the same argument to $-\phi$ for a local maximum.

Second-order necessary condition: multi-dimensional cases

$\mathbf{A} \succeq 0$: PSD; $\mathbf{A} \succ 0$: PD; $\mathbf{A} \preceq 0$: NSD; $\mathbf{A} \prec 0$: ND.

Theorem 6.5 (Second-order necessary condition)

Let $f \in C^2$ near an interior point $\mathbf{x}^*$.

- Local minimum $\Rightarrow \nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$.
- Local maximum $\Rightarrow \nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \preceq 0$.

Indeed, for every direction $\mathbf{d}$,

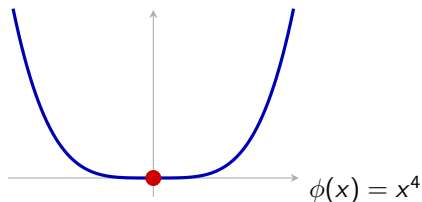
$$\phi_{\mathbf{d}}''(0) = \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d}.$$

Example 6.6

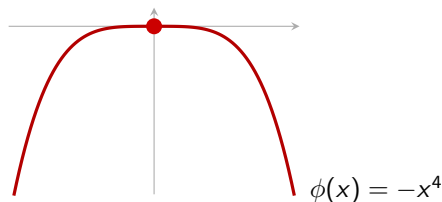
Second-order conditions tell x^2 from $-x^2$, $x^2 - x^4$ from $-x^2 + x^4$.

A zero Hessian can hide global extrema

Both functions satisfy $\phi'(0) = \phi''(0) = 0$.



Role of 0: global minimum

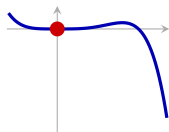


Role of 0: global maximum

The quadratic model is flat; the fourth-order term decides.

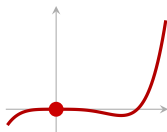
The same zero Hessian allows three more roles

All three functions also satisfy $\phi'(0) = \phi''(0) = 0$.



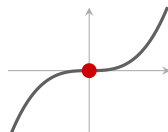
$$\phi(x) = x^4 - x^5$$

**Role of 0: local minimum,
not global**



$$\phi(x) = -x^4 + x^5$$

**Role of 0: local maximum,
not global**

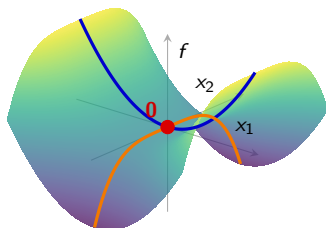


$$\phi(x) = x^3$$

Role of 0: saddle point

The same first- and second-order data lead to five different roles.

A singular PSD Hessian can still hide a saddle



$$f(x_1, x_2) = x_1^2 - x_2^4.$$

$$\nabla f(\mathbf{0}) = \mathbf{0},$$

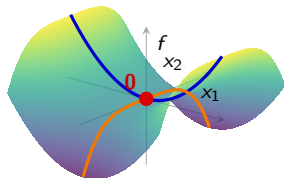
$$\mathbf{H}_f(\mathbf{0}) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \succeq 0.$$

Two cross-sections

$$f(t, 0) = t^2 > 0 \quad f(0, t) = -t^4 < 0.$$

$\Rightarrow$ The origin is a saddle point.

Blind spot in the Taylor expansion



At a stationary point,

$$f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) = \frac{1}{2} \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + o(\|\mathbf{d}\|^2).$$

If $\mathbf{H}_f(\mathbf{x}^*)$ is singular, then for some $\mathbf{d} \neq \mathbf{0}$,

$$\mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} = 0.$$

Along these zero- or near-zero-curvature directions, the quadratic term has no uniform positive margin. Higher-order terms can decide the sign.

Convexity supplies the missing comparison

For a differentiable convex function on a convex domain,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \forall \mathbf{x}, \mathbf{y}.$$

At a stationary point $\mathbf{x}^*$, $\nabla f(\mathbf{x}^*) = 0$,

$$f(\mathbf{y}) \geq f(\mathbf{x}^*) \quad \forall \mathbf{y}.$$

Why convexity is different

This is an exact inequality over the whole domain, not a local Taylor truncation.

Necessary conditions are filters, not certificates

For an interior local minimum,

$$\text{local minimum} \implies \begin{cases} \nabla f(\mathbf{x}^*) = \mathbf{0}, \\ \mathbf{H}_f(\mathbf{x}^*) \succeq 0. \end{cases}$$

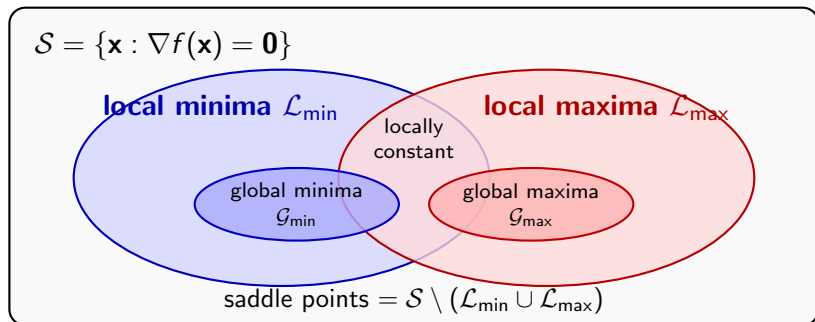
- Failure of a necessary condition rejects a candidate.
- Passing both conditions does not certify a minimum.
- A semidefinite Hessian leaves zero-curvature directions unresolved.

Candidate enumeration is a conditional workflow

- 1 Identify the problem at hand is **unconstrained**.
- 2 (Establish that **a global minimizer exists.**)

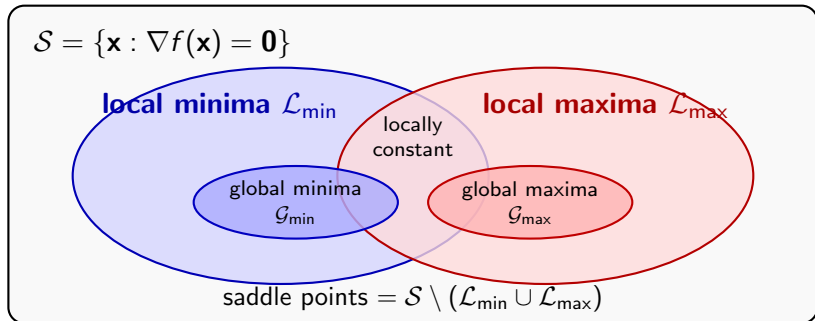
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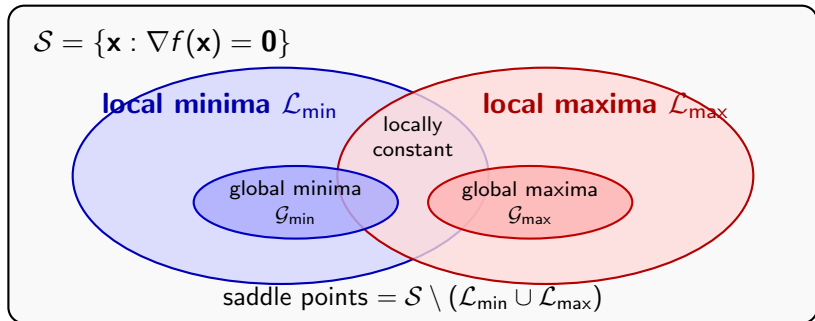
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- 4 If the stationary set is **finite**, compare objective values (to decide global optimality).



Candidate enumeration is a conditional workflow

- 1 Identify the problem at hand is **unconstrained**.
- 2 (Establish that **a global minimizer exists.**)
- 3 Solve $\nabla f(\mathbf{x}) = \mathbf{0}$ for all stationary points.
- 4 If the stationary set is **finite**, compare objective values (to decide global optimality).
- 5 Classify candidates with $\mathbf{H}_f(\mathbf{x})$ when possible (more in next lecture).



Example: candidate search by necessary condition

Example 6.7

Consider

$$f(x_1, x_2) = (x_1^2 - 1)^2 + (x_2 - 2)^2.$$

- Show that f attains its global minimum.
- Enumerate all the stationary points of f .
- Determine the global minimizer(s) of f .

(i) Unconstrained optimization. f is continuous and coercive, so a global minimizer exists.

The stationary equations are polynomial and have finitely many solutions.

Example: candidate search by necessary condition (cont'd)

$$f(x_1, x_2) = (x_1^2 - 1)^2 + (x_2 - 2)^2.$$

(ii) Enumerate all the stationary points of f .

The gradient is

$$\nabla f(x_1, x_2) = \begin{bmatrix} 4x_1(x_1^2 - 1) \\ 2(x_2 - 2) \end{bmatrix}.$$

Hence

$$x_1 \in \{-1, 0, 1\}, \quad x_2 = 2.$$

The complete stationary set is

$$\mathcal{S} = \{(-1, 2), (0, 2), (1, 2)\}.$$

Example: candidate search by necessary condition (cont'd)

$$f(x_1, x_2) = (x_1^2 - 1)^2 + (x_2 - 2)^2.$$

(iii) Determine the global minimizer(s) of f .

$$f(-1, 2) = 0, \quad f(0, 2) = 1, \quad f(1, 2) = 0.$$

Therefore

$$\arg \min_{\mathbf{R}^2} f = \{(-1, 2), (1, 2)\}.$$

$(0, 2)$ must not be, because $\mathbf{H}_f((0, 2)) = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix} \not\geq 0$.

Example: candidate search by necessary condition (cont'd)

$$f(x_1, x_2) = (x_1^2 - 1)^2 + (x_2 - 2)^2.$$

(iii) Determine the global minimizer(s) of f .

$$f(-1, 2) = 0, \quad f(0, 2) = 1, \quad f(1, 2) = 0.$$

Therefore

$$\arg \min_{\mathbf{R}^2} f = \{(-1, 2), (1, 2)\}.$$

$(0, 2)$ must not be, because $\mathbf{H}_f((0, 2)) = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix} \not\geq 0$. In fact, it is a saddle point:

$$f(t, 2) = (t^2 - 1)^2 < f(0, 2) \quad \forall 0 < |t| < \sqrt{2};$$

$$f(0, 2 + t) = 1 + t^2 > f(0, 2) \quad \forall t \neq 0.$$

Trap 1: unconstrained vs constrained

The derivative need not vanish at a boundary optimizer:

$$\min_{x \in [0,1]} \phi(x) = x.$$

The global minimizer is $x^* = 0$, but $\phi'(0) = 1$.

- At an interior point, any perturbative direction is admissible.
- At a boundary point, the feasible motion may be constrained.

Trap 2: finite stationarity set $\nRightarrow$ well-posedness

Consider

$$\phi(x) = x^3 - 3x.$$

Its stationary points are $x = \pm 1$, with

$$\phi(1) = -2 < 2 = \phi(-1).$$

Choosing $x = 1$ by comparison would be wrong:

$$\phi(x) \rightarrow -\infty \quad \text{as } x \rightarrow -\infty.$$

But a not-well-posed problem can still have local minimizers/maximizers or saddle points.

Trap 3: infinitely many strict local minima

$$f(x) = \sin^2 x + 1 - \cos x,$$

$$f'(x) = \sin x(2 \cos x + 1),$$

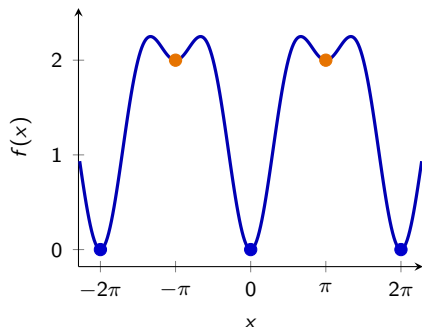
$$f''(x) = 2 \cos(2x) + \cos x.$$

for every $k \in \mathbb{Z}$,

$$f'(k\pi) = 0, \quad f''(k\pi) = 2 + (-1)^k > 0.$$

Hence every $k\pi$ is a stationary point:

- $f(2k\pi) = 0$: global minima;
- $f((2k+1)\pi) = 2$: local, not global.



• global minima • local, not global

Even isolated strict local minima need not form a finite checklist!

Take-home messages

- An interior local minimizer: $\nabla f(\mathbf{x}^*) = 0$, $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$.
- These necessary conditions filter candidates but do not certify them.
- Checklist before certifying candidates:
 - unconstrained?
 - (for global optimality) well-posed?
 - finitely many?
- **Question:** how to certify the stationary point?

Reference appendix: proof map

The classroom argument used one-dimensional restrictions:

$$\mathbf{x}^* \text{ locally optimal} \implies t = 0 \text{ locally optimal for } f(\mathbf{x}^* + t\mathbf{d}).$$

The reference slides record the full steps for:

- 1 the multivariate first-order condition;
- 2 the multivariate second-order condition; and
- 3 the finite stationary-point search.

Reference: proof of the multivariate first-order condition

Since $\mathcal{D}$ is open, for every $\mathbf{d} \in \mathbf{R}^n$ the restriction

$$\phi_{\mathbf{d}}(t) = f(\mathbf{x}^* + t\mathbf{d})$$

is defined for all sufficiently small positive and negative t . If $\mathbf{x}^*$ is a local minimum, then 0 is a local minimum of $\phi_{\mathbf{d}}$. The one-dimensional result gives

$$0 = \phi'_{\mathbf{d}}(0) = \nabla f(\mathbf{x}^*)^T \mathbf{d} \quad \forall \mathbf{d}.$$

Choosing $\mathbf{d} = \nabla f(\mathbf{x}^*)$ yields $\|\nabla f(\mathbf{x}^*)\|^2 = 0$, hence $\nabla f(\mathbf{x}^*) = \mathbf{0}$. The maximum case follows by applying the result to $-f$.

Reference: proof of the second-order condition

Let $\mathbf{x}^*$ be a local minimum and fix $\mathbf{d} \in \mathbf{R}^n$. For $\phi_{\mathbf{d}}(t) = f(\mathbf{x}^* + t\mathbf{d})$, the first-order result gives $\phi'_{\mathbf{d}}(0) = 0$. Taylor expansion at 0 gives

$$\phi_{\mathbf{d}}(t) - \phi_{\mathbf{d}}(0) = \frac{1}{2}t^2\phi''_{\mathbf{d}}(0) + o(t^2).$$

If $\phi''_{\mathbf{d}}(0) < 0$, the right-hand side is negative for all sufficiently small nonzero t , contradicting local minimality. Hence

$$\mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} = \phi''_{\mathbf{d}}(0) \geq 0 \quad \forall \mathbf{d},$$

so $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$. Apply the result to $-f$ for a local maximum.